

Clustering Algorithms

① K Means Algorithm

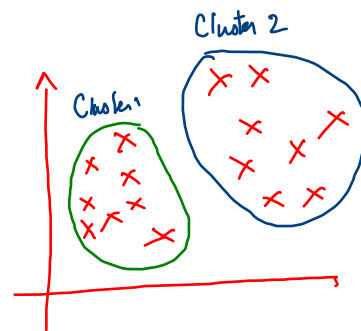
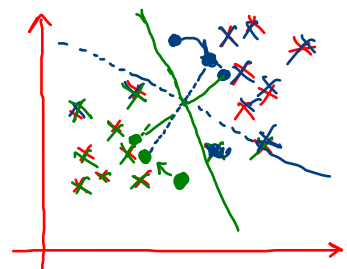
② Hierarchical

③ DBSCAN

Agglomerative Divisive

① K Means

Step 1 → Initialise k centroids
 $k=2$

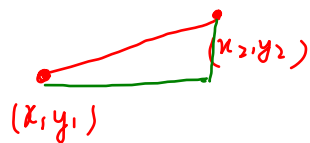


Step 2 → Cluster data points near to centroid in one cluster

Step 3 → Move the centroids towards mean.

Step 4 → Repeat until new data points are not clustered

Distance

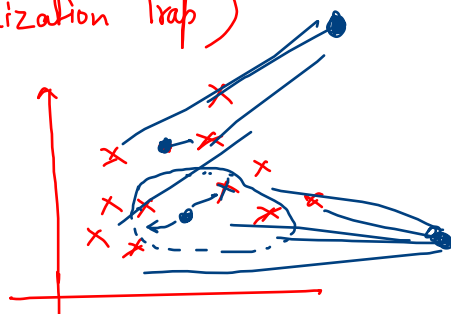


Euclidean distance = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Manhattan distance = $|y_2 - y_1| + |x_2 - x_1|$

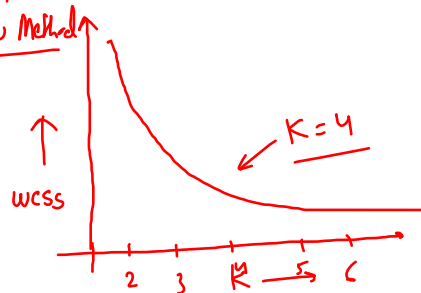
K-Means Challenge (Random Initialization Trap)

Fix - K-Means ++ initialization



***** How to select optimal K - Elbow Method

WCSS → Within cluster sum of squares



$k=2$



⇒ 850

$k=3$



⇒ 750

② Hierarchical Clustering

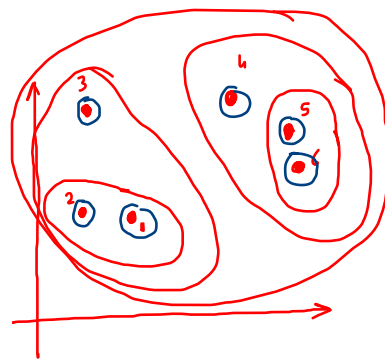
- Agglomerative
- Divisive

Agglomerative

Step 1 → Assign each data point individual clusters

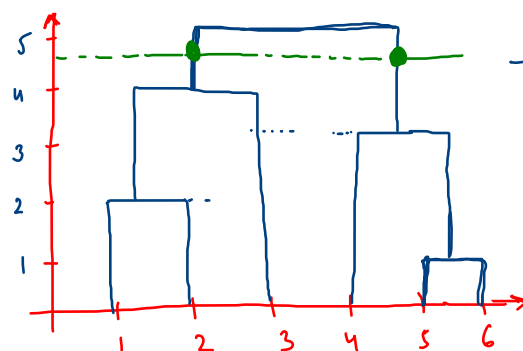
Step 2 → Assign closest data points into one cluster

Step 3 → Repeat step 2 until all data points are in one cluster.



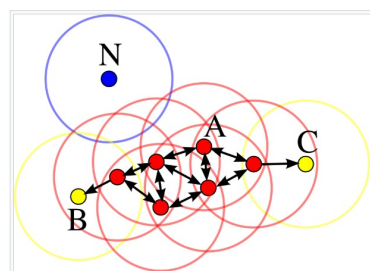
Dendrogram

no. of clusters = 2



Find longest vertical line not intersected by any horizontal line

③ DBSCAN - Density Based Spatial Clustering Application with noise



① Core points -

② Border points -

③ Noise/outlier -

$A \rightarrow B \rightarrow C$

$A \rightarrow C$

Parameters

Min. pts

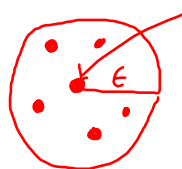
Epsilon

Core pt.

Min. pts = 4

$1 < \text{pt} < \text{min. pts}$

Border point



Zero points

Noise/outlier

Works well with Non-Linear Data

Silhouette Score

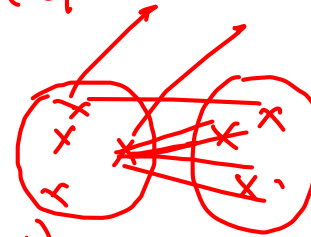
For data point $i \in C_i$ (data point i in the cluster C_i), let

$$a(i) = \frac{1}{|C_i| - 1} \sum_{j \in C_i, i \neq j} d(i, j)$$

be the mean distance between i and all other data points in the same cluster, where $|C_i|$ is the number of points belonging to cluster C_i , and $d(i, j)$ is the distance between data points i and j in the cluster C_i (we divide by $|C_i| - 1$ because the distance $d(i, i)$ is not included in the sum). $a(i)$ can be interpreted as a measure of how well i is assigned to its cluster (the smaller the value, the better the assignment).

$$\textcircled{1} \quad \underline{a(i)} = \frac{1}{|C_i| - 1} \sum_{j \in C_i} d(i, j)$$

$$\textcircled{2} \quad b(i) = \min_{j \neq i} \frac{1}{|C_j|} \sum_{j \in C_j} d(i, j)$$



For each data point $i \in C_i$, we now define

$$b(i) = \min_{j \neq i} \frac{1}{|C_j|} \sum_{j \in C_j} d(i, j)$$

to be the *smallest* (hence the **min** operator in the formula) mean distance of i to all points in any other cluster (i.e., in any cluster of which i is not a member). The cluster with the smallest mean dissimilarity is said to be the "neighboring cluster" of i because it is the next best fit cluster for point i .

We now define a *silhouette* (value) of one data point i

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}, \text{ if } |C_i| > 1$$

and

$$s(i) = 0, \text{ if } |C_i| = 1,$$

which can also be written as

$$s(i) = \begin{cases} 1 - \frac{a(i)}{b(i)}, & \text{if } a(i) < b(i) \\ 0, & \text{if } a(i) = b(i) \\ \frac{b(i)}{a(i)} - 1, & \text{if } a(i) > b(i) \end{cases}$$

$\textcircled{3}$ Calculate silhouette score

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

if $a(i) > b(i)$

$$s(i) = \frac{b(i) - a(i)}{a(i)} = \frac{b(i)}{a(i)} - \frac{a(i)}{a(i)} = \frac{b(i)}{a(i)} - 1$$

$$s(i) \rightarrow \underline{-1 \text{ tol}}$$

1 - perfect cluster

-1 - Bad cluster